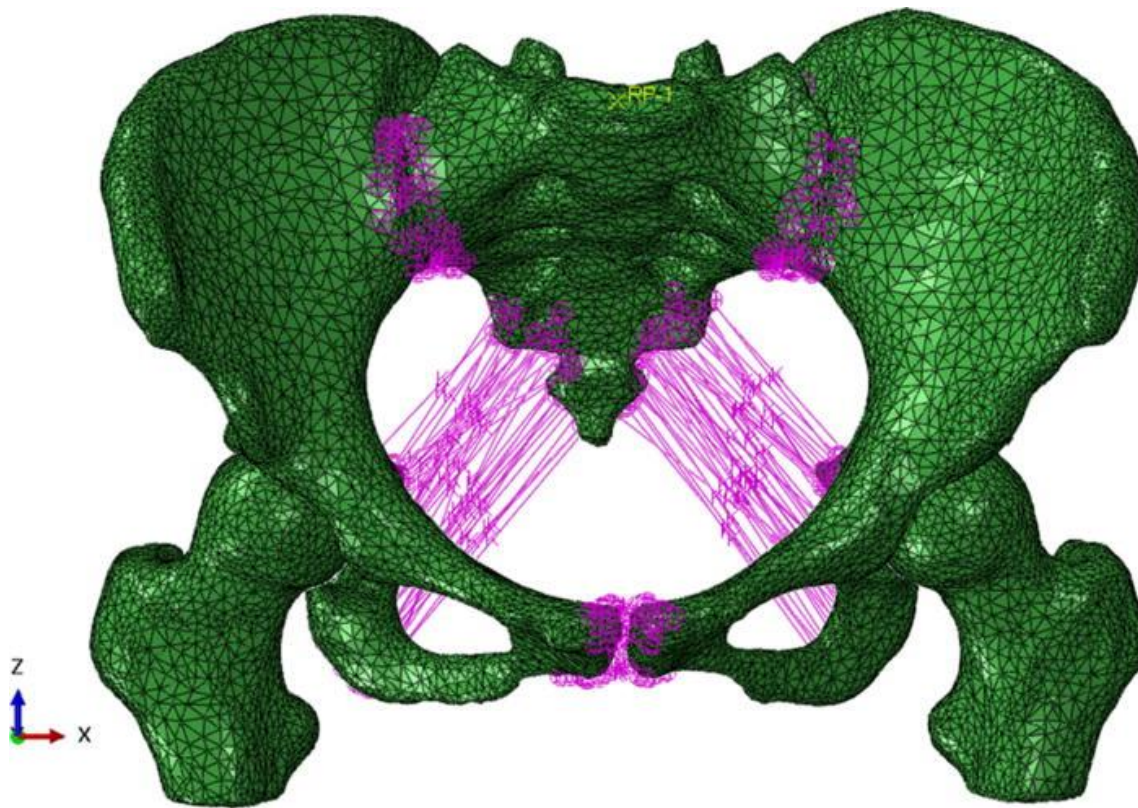


Lecture 15 – Modern Design Tools

James Busfield



Report Submission Schedule

A report including all your team's supporting evidence. Submitted electronically before 11:00pm 17th March 2025 (today) using the coursework upload facility on QM+ under the assessment tab.

 **Group Case Study Report - Submission Point EMS501U**

Drag your blocks to here.

[Assignment](#) [Settings](#) [Advanced grading](#) [Download all submissions \(username and ID\)](#) [More ▾](#)

[Make a submission](#)

Opened: Friday, 14 March 2025, 12:00 AM
Due: Monday, 17 March 2025, 11:00 PM

Please submit your group report on this portal. Only one member from each team should submit the report per group.

The deadline to submit the group report for all groups is 17th March 2024 at 23:00.

[View all submissions](#) [Grade](#)

Grading summary

The report format must reflect the needs of your case study. One single format is not suitable for all six case studies. You should include the following though:

- Design goals / project brief – Briefly identify the function of the product behaviour or properties that you consider the most important
- Constraints, objectives and free variables. Translate the design brief into specific materials properties that are optimal for your application
- List suitable materials solutions that can be used with justification. Rank them and identify your optimal solution. Cross-reference to known patents, papers or products on the market already.
- Processing - Identify the optimal way of making your product for this specific application - justify your choices.
- Summary - Provide a brief summary of materials and processing options that you have gone for
- References

Report Rules New for this Year

- Option 1 – Submit a report that is 4 pages or less without including cover and references
- Option 2 - A **maximum** word count of 2500 words. If you opt for this then please include your word count on the cover page. Do not have to include the references and the cover page in your word count.
- Cover Page must include (Not in the word or page count)
 - The title
 - Student names and ID numbers
 - A short abstract (ideally less than 100 words) explaining your chosen solution
 - Any visuals / style formatting that you desire.

Remember to use a Decision Matrix

Decision matrix

[Table 2: Property importance comparison]

	Yield Strength	Elastic modulus	Corrosion resistance	Price	Density	Sustainability	Total Weight
Yield Strength		1	1	1	1	1	5
Elastic Modulus	0		1	1	1	1	4
Corrosion Resistance	0	0		1	0	1	2
Price (m ³)	0	0	0		0	1	1
Density (kg/m ³)	0	0	1	1		1	3
Sustainability	0	0	0	0	0		0

Material Performance index

To find the material index, the main parameters need to be compared to be taken into consideration. Firstly, the material needs to have a high strength to weight ratio, coupled with other important parameters like corrosion resistance and elastic modulus. The formula, as displayed below, illustrates the criteria necessary for selecting the appropriate material. Although secondary parameters such as cost and sustainability are important, the primary parameters must be prioritized and adjusted accordingly depending on their weighting, requiring some compromise. Density was given high importance due to its interconnection with sustainability and price because of its significant effect on fuel economy. Multiplying properties in the material index linearly depending on their weighting yielded satisfactory results. For example, density would be multiplied by a coefficient of 3.

$$M = \frac{\text{Young's Modulus} \times \text{Yield Strength}}{\text{Density per unit area} \times \text{Cost}} = \frac{\sigma_y \times E}{\rho \times C_m}$$

Potential materials

Case Study - Presentation Guidelines

- Preparation is vital / rehearse the schedule
- Know your script
- No more than 2 presenters (pick your best team)
- 7 minutes is not long - Keep to time
- 3 words per second > 1400 words maximum
- 7 pages of a PPT maximum
- Do not read a script or for that matter words on your presentation
- Makes sure that the presentation supports your report
- Keep fonts size above 20pt – this is 20 point this is 12 point this is 6 point this is 3 pt
- Think about your target audience

This week

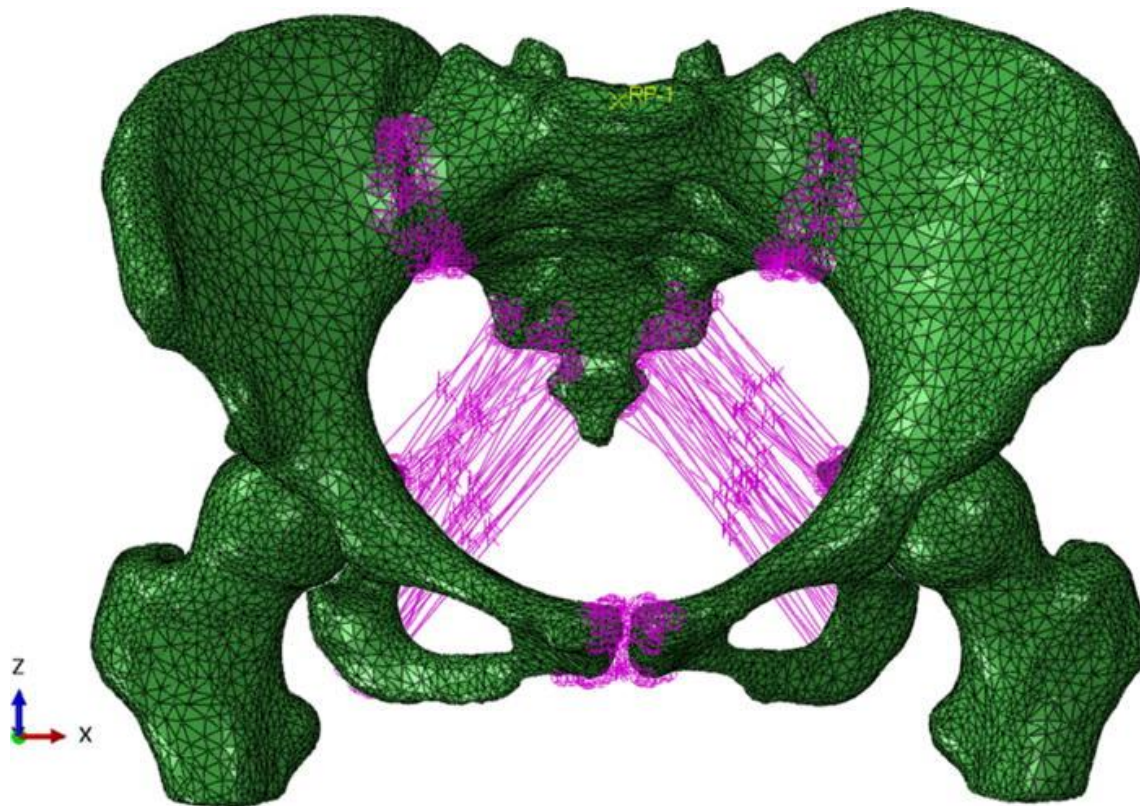
17-Mar-25	Monday	10:00 - 11:00	Lecture 1A	Skeel LT	JB	L15: Design Tools
17-Mar-25	Monday	13:00 - 14:00	Lecture 1B	Eng 209	JB/SRJ	L15: Design Tools
17-Mar-25	Monday	14:00 - 15:00	IT Group A	QB 202 PC Lab	Teaching Team	Quiz 7: Did you master Edupack Software
17-Mar-25	Monday	15:00 - 16:00	IT Group B	QB 202 PC Lab	Teaching Team	Quiz 7: Did you master Edupack Software
17-Mar-25	Monday	16:00 - 17:00	IT Group C	QB 202 PC Lab	Teaching Team	Quiz 7: Did you master Edupack Software
17-Mar-25	Monday	17:00 - 18:00	IT Group D	QB 202 PC Lab	Teaching Team	Quiz 7: Did you master Edupack Software
17-Mar-25	Monday	23:00	Online QM+ Submission			Submission of Case Study Reports
20-Mar-25	Thursday	09:00 - 10:00	Sem 1	Eng 209	JB/SRJ	Feedback Week Quiz 7 + Kahoot
20-Mar-25	Thursday	14:00 - 15:00	Sem 2	Fogg LT	JB/SRJ	Feedback Week Quiz 7 + Kahoot
20-Mar-25	Thursday	15:00 - 16:00	Sem 3	G. O. Jones LT	JB/SRJ	Feedback Week Quiz 7 + Kahoot
21-Mar-25	Friday	17:00 - 18:00	Lecture 2	Arts Two LT	JB/SRJ	L16: Exam Preparation

Next Week

24-Mar-25	Monday	10:00 - 11:00	Lecture 1A	Skeel LT	JB/SRJ	Revision Session
24-Mar-25	Monday	13:00 - 14:00	Lecture 1B	Eng 209	JB/SRJ	Revision Session
24-Mar-25	Monday	14:00 - 15:00	IT Group A	QB 202 PC Lab	Teaching Team	Quiz 8: Mock Exam 1
24-Mar-25	Monday	15:00 - 16:00	IT Group B	QB 202 PC Lab	Teaching Team	Quiz 8: Mock Exam 1
24-Mar-25	Monday	16:00 - 17:00	IT Group C	QB 202 PC Lab	Teaching Team	Quiz 8: Mock Exam 1
24-Mar-25	Monday	17:00 - 18:00	IT Group D	QB 202 PC Lab	Teaching Team	Quiz 8: Mock Exam 1
26-Mar-25	Wednesday	10:00 - 12:30	Presentation	Queens LG1	SJ	Presentation Practice group Meeting CS6
26-Mar-25	Wednesday	13:00 - 15:30	Presentation	GO Jones UG1	RS	Presentation Practice group meeting CS2
27-Mar-25	Thursday	09:00 - 10:00	Sem 1	Eng 209	JB/RS	Presentations CS6
27-Mar-25	Thursday	14:00 - 15:00	Sem 2	Fogg LT	JB/RS	Presentations CS6
27-Mar-25	Thursday	15:00 - 16:00	Sem 3	G. O. Jones LT	JB/RS	Feedback CS6
28-Mar-25	Friday	17:00 - 18:00	Lecture 2	Arts Two LT	SRJ	Feedback on Quiz 8: Mock Exam 1

Lecture 15 – Using Modern Design Tools

Prof James Busfield



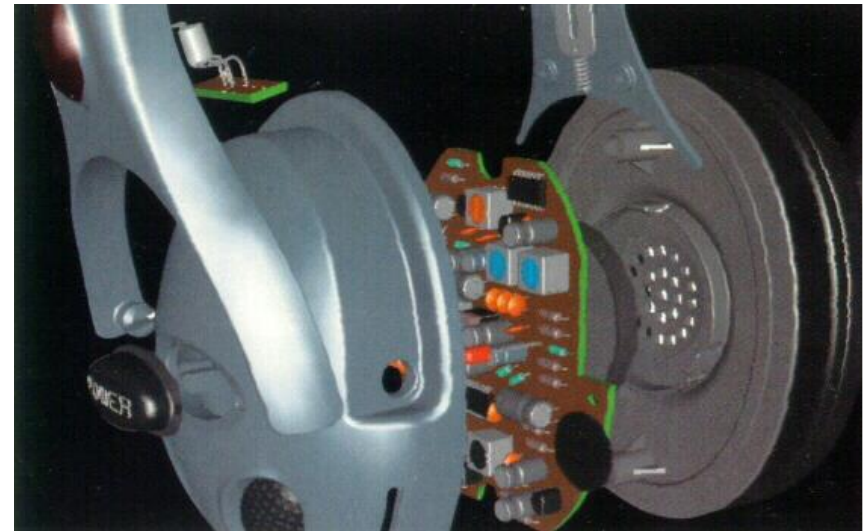
Modern manufacturing relies heavily on computer systems to address the needs for:

- high-quality
- reliable
- usable
- cost-effective products

The purpose of this lecture is to explain the use of computer based design tools



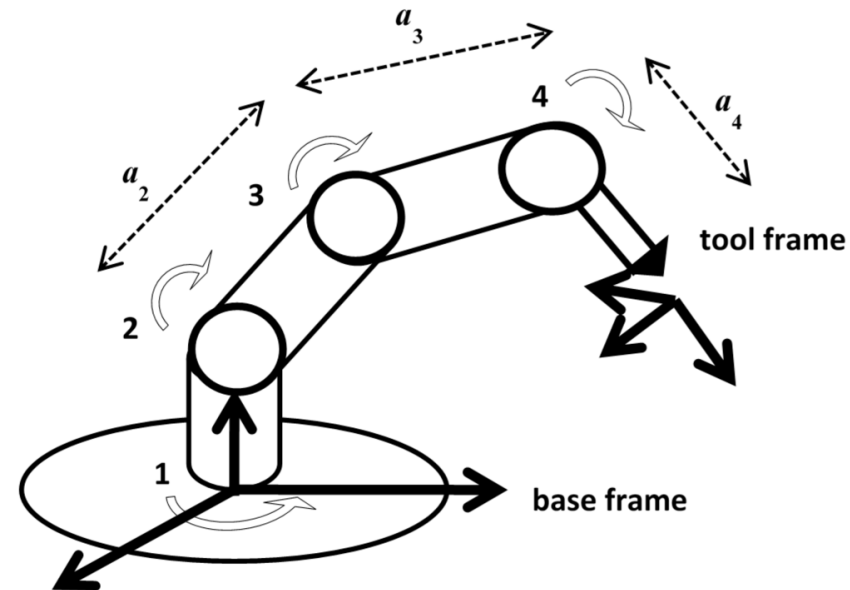
- Using CAE - a part description is not a picture, but rather a graphic representation of co-ordinate points and geometric shapes from which a picture can be constructed.
- Parts can be:
 - drawn
 - rotated
 - scaled and
 - presented in 3D
- It enables easy visualisation
- It also allows for:
 - repetitive sections to be redrawn automatically
 - overlays to be easily shown on screen
 - engineering drawings to be rapidly updated and printed
 - tool-cutting paths for structures to be programmed



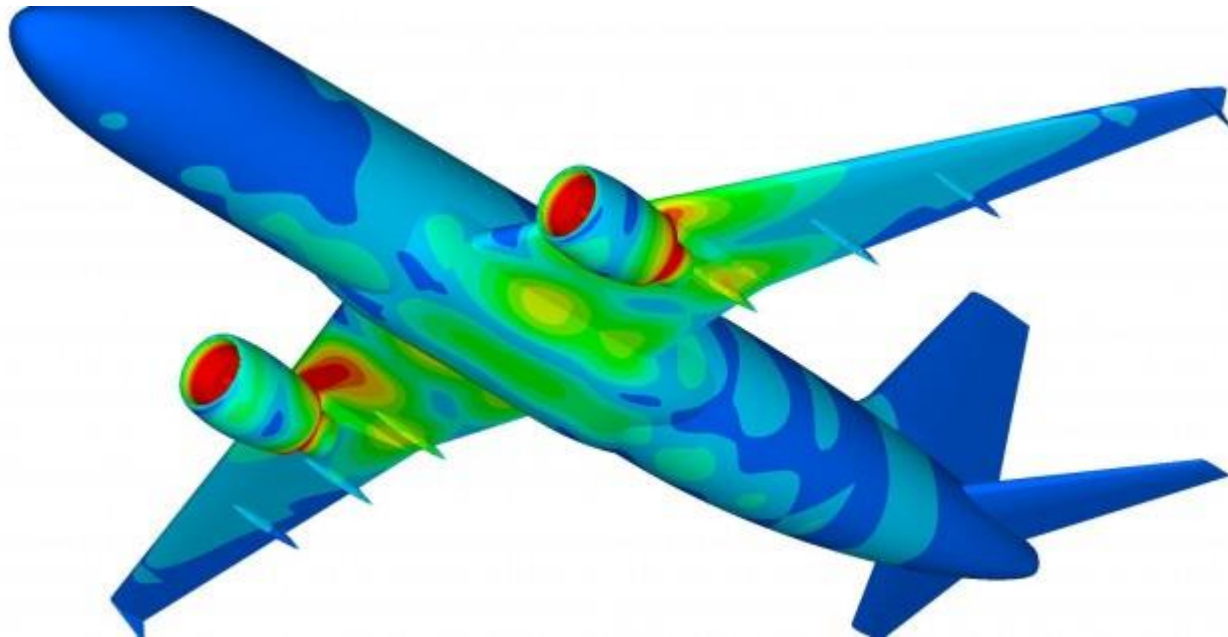
- In geometric modelling, the designer constructs the shape of the object on the computer screen
- This pictorial representation is converted into an analytical model by the computer, which stores the information in a database
- Originally 3D models were wire frames that represented a parts shape as a series of interconnecting line element
- Now sophisticated systems work with a full 3D geometric model of the part



- The kinematic features of CAD systems permits the animated motion of hinged parts such as front-end loader buckets or aircraft landing gear to be evaluated
- This allows the designer to ensure that the moving parts neither interfere with one another or impact on adjoining parts of a structure
- Clearances and positions can be checked in the early design stages
- Thus potential fit problems can be eliminated



- Most CAD systems allow the user to move directly from a geometric model to engineering analysis.
- Information such as moment of inertia, volume, mass, or projected areas can be easily calculated.
- The most powerful analytical tool include finite element analysis and computational fluid mechanics



- Finite Element Analysis (FEA) is used to solve engineering problems
- It is widely used in industry
- It has made many numerical analysis techniques and experimental testing methods redundant.
- For example, in the car industry, the structural integrity and performance of any new car design is thoroughly analysed and evaluated with finite element models, typically a year before a first prototype is built.
- The technique is adopted to analyse for example:
 - The strength of components and that of the car as a whole.
 - Its impact behaviour and crash worthiness.
 - The frequency characteristics of the total structure and its separate components.
 - The temperature distribution in the engine block and pistons and the resulting thermal stresses.

- **Steady state problems**

- These analyses are the most common use of the finite element method.
- For elasticity problems, a body under equilibrium conditions can be analysed and its distortion predicted.
- From the calculated values of displacement, it is then possible to derive the strains and stresses experienced by the body.
- Can be thermal or structural problems

- **Eigenvalue Problems**

- Eigenvalue problems are an extension of equilibrium problems to determine the natural frequencies and mode shapes of components, and the buckling loads of structures

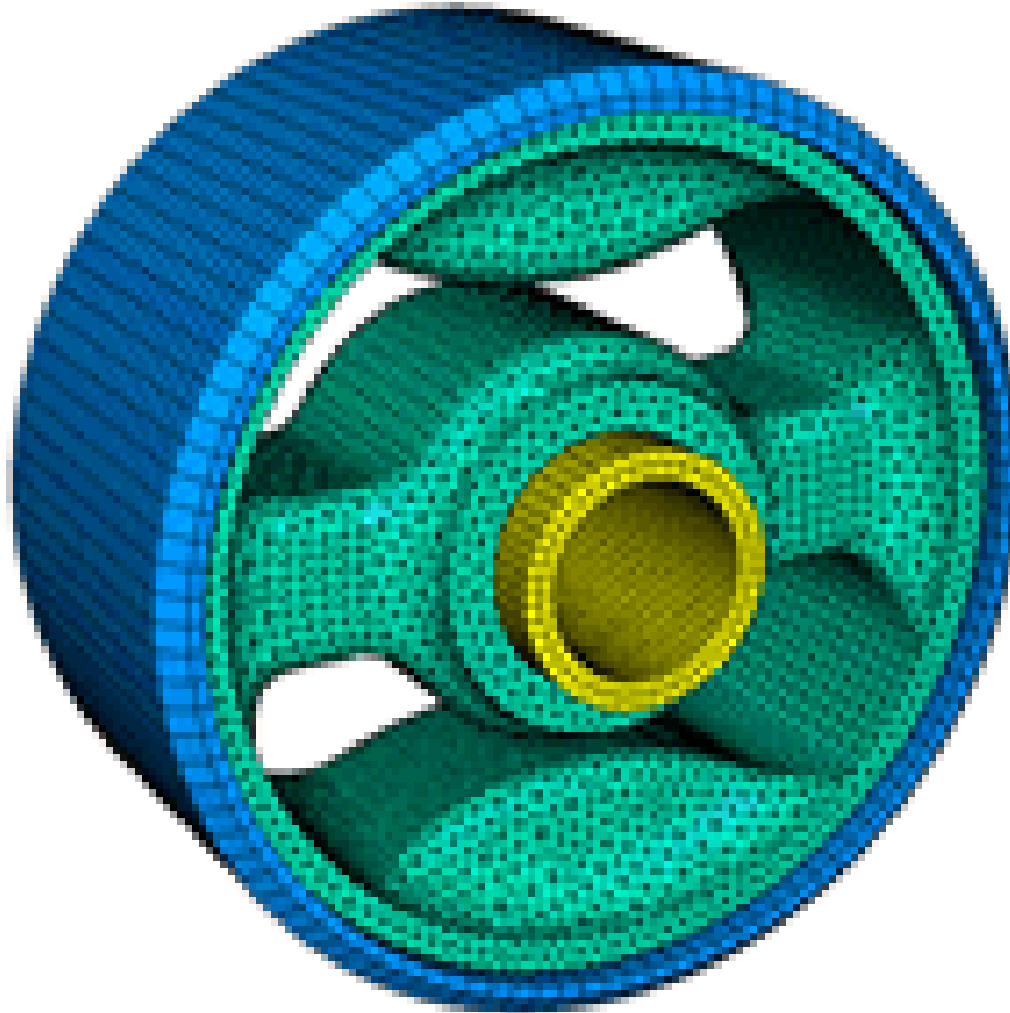
- **Transient problems**

- In transient or propagation problems the loads can be functions of time
- The finite element method is used to calculate the forced response of the body

Steady State Analysis

Area	Typical application
Aerospace engineering	Stress analysis of aircraft frames, wings, missile and spacecraft components; thermal analysis of gas turbine blades, heat exchangers.
Automotive engineering	Stress analysis of crankshaft, cylinder block, connecting rods, chassis; thermal analysis of pistons; lubrication of big-end bearings.
Biomedical engineering	Stress analysis of bones, hip replacements, teeth and heart.
Civil engineering	Stress analysis of dams, retaining walls, excavations; soil mechanics.
Electrical engineering	Steady state thermal analysis of integrated circuit boards.
Mechanical engineering	General one-, two- and three-dimensional and axisymmetric stress analyses of components; stress analysis of shafts, gears and pressure vessels; crack propagation.
Nuclear engineering	Stress analysis of reactor vessels and structures; thermal analysis of reactor components.

Typical steady state model



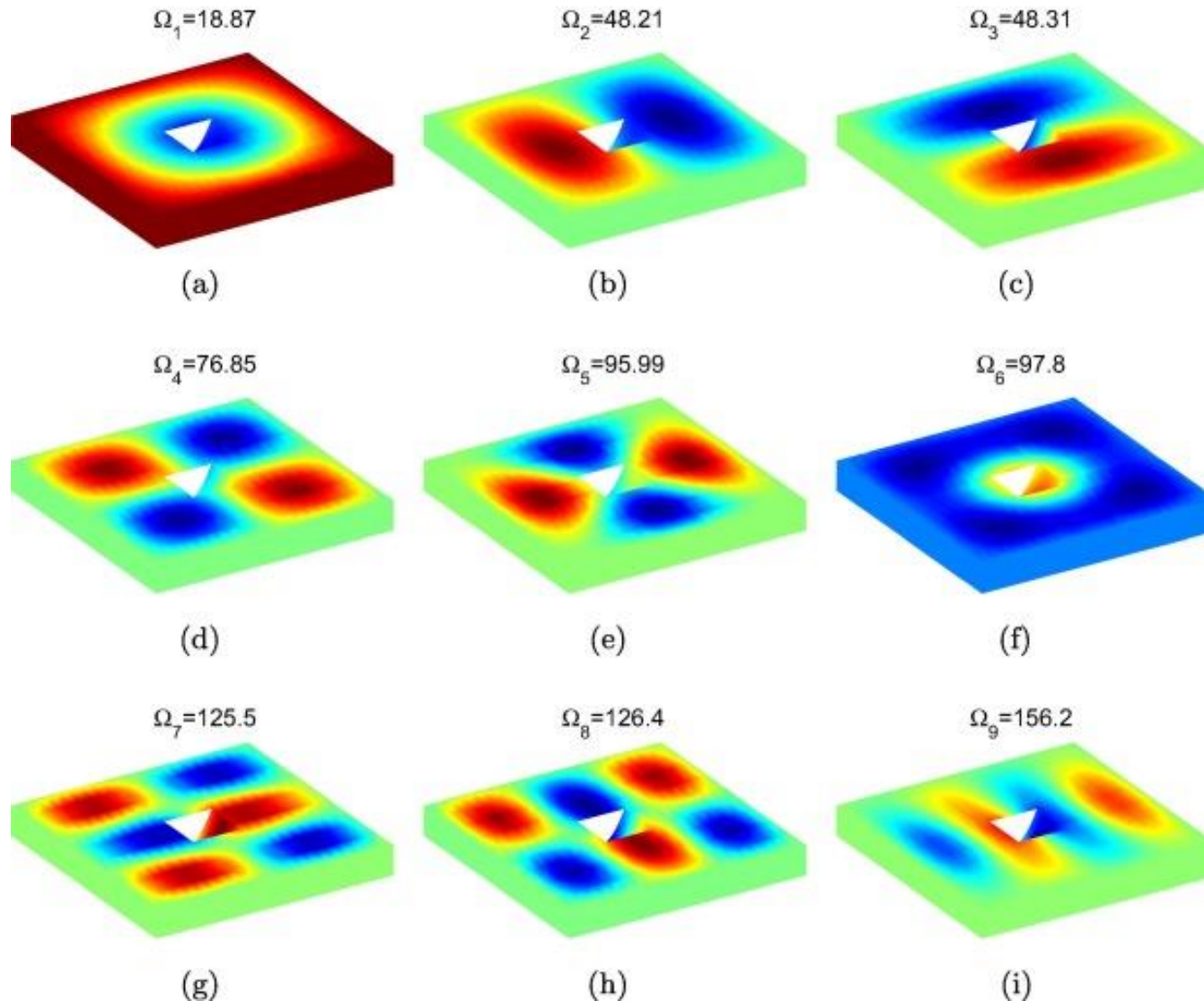
Eigenvalue Analysis

Area	Typical application
Aerospace engineering	Frequency analysis of engine components, helicopter rotor blades, gearbox casing; acoustic analysis of aircraft passenger compartments
Automotive engineering	Acoustic analysis of passenger compartment and exhaust system; frequency analysis of gearbox casing and body shell
Electrical engineering	Natural frequencies of printed circuit boards
Hydraulic engineering	Natural periods of lakes and harbours; sloshing of liquids
Mechanical engineering	Natural frequency of components, shafts; critical buckling loads
Nuclear engineering	Neutron flux distribution; frequency analysis of pressure vessels
Structural engineering	Natural frequency and buckling loads of structures; vibration analysis of multi-storey buildings

Modal Response - Tacoma Narrows Bridge



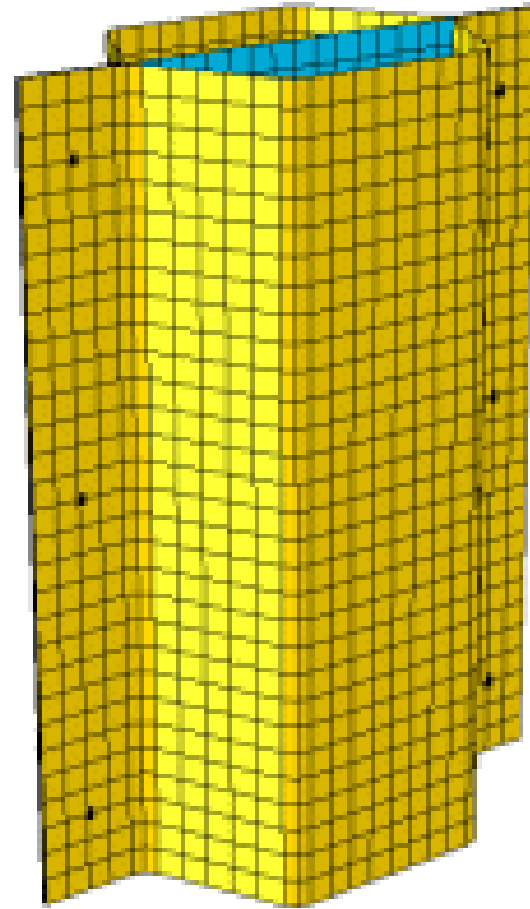
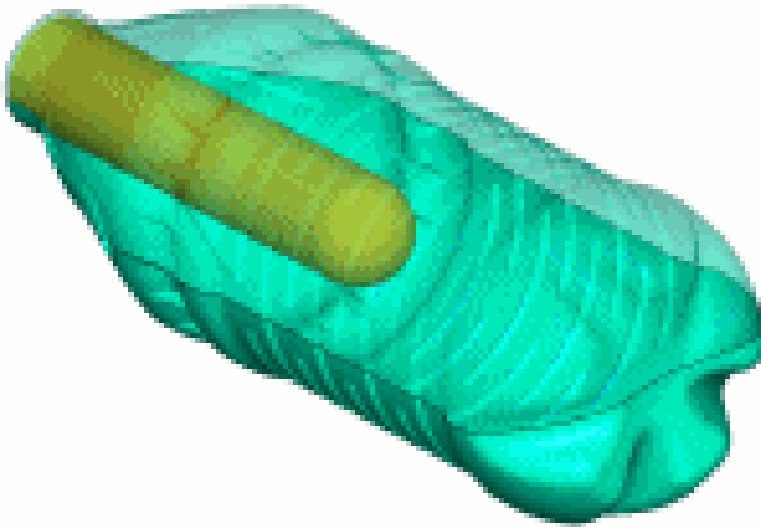
Modelling of Resonance



Transient Analysis

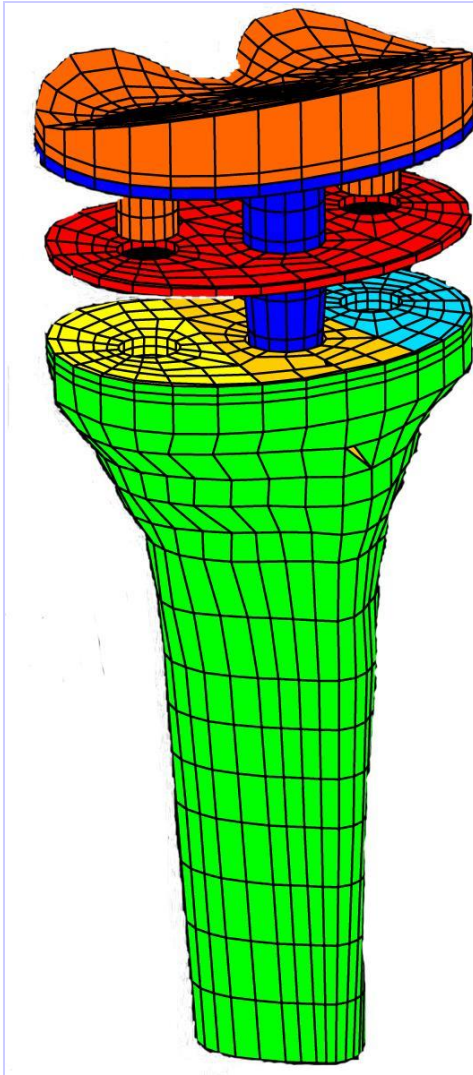
Area	Typical application
Aerospace engineering	Forced frequency analysis of aircraft and spacecraft components thermal analysis of rocket nozzles
Automotive engineering	Time dependent analysis of engine components, piston, disc brakes, exhaust; crash worthiness of chassis
Biomedical engineering	Impact analysis of skull; dynamic analysis of body and limbs
Civil engineering	Stress waves in rock structures
Mechanical engineering	Analysis of impact problems; dynamic crack propagation
Nuclear engineering	Time dependent thermal analysis of reactor components; shock spectrum analyses
Structural engineering	Shock and earthquake analysis of buildings and bridges

Transient (Explicit) Dynamic Analysis

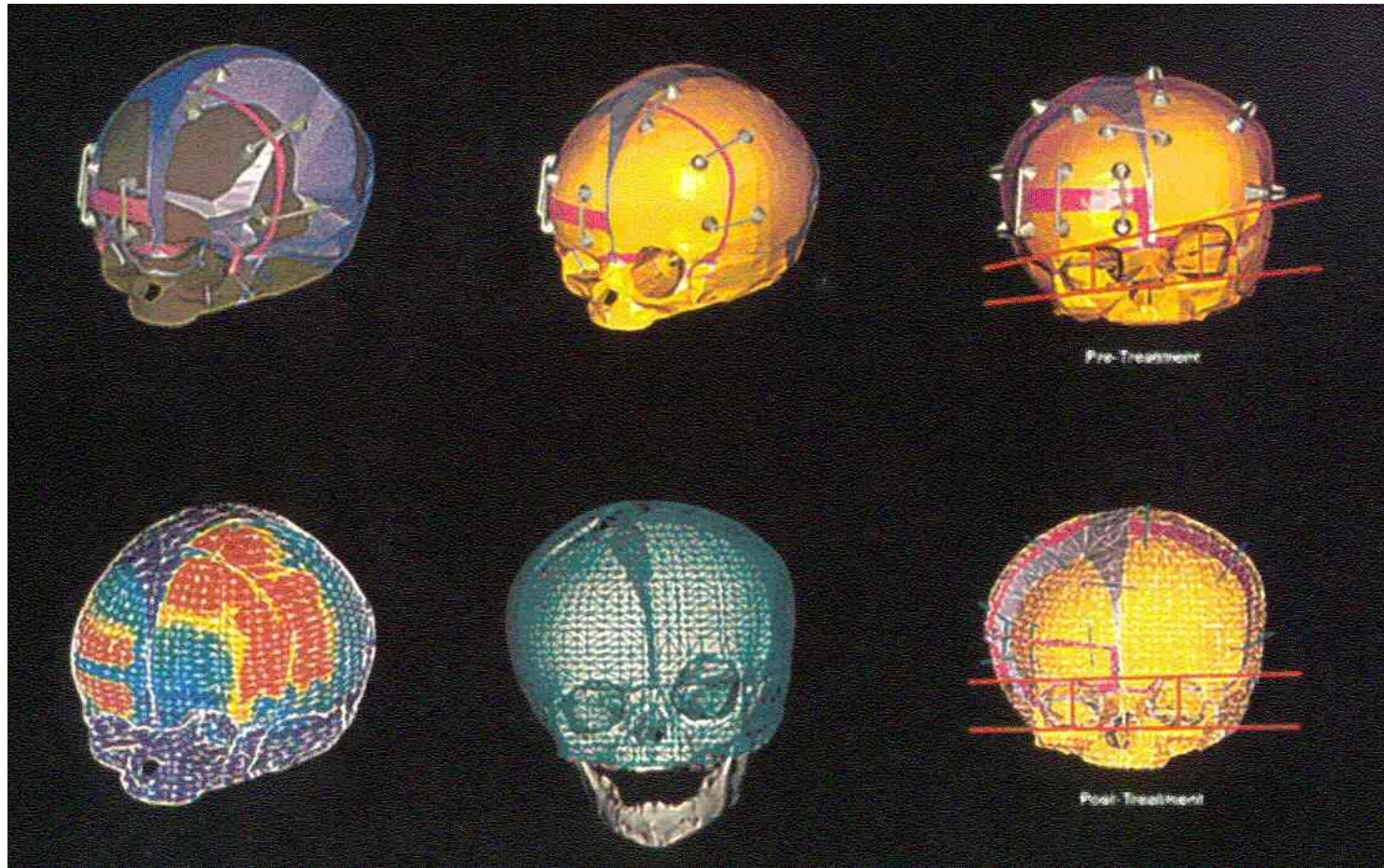


- Consider a body in which the distribution of an unknown variable (such as displacement) is required.
- The region is divided into an assembly of subdivisions called elements, which are considered to be interconnected at points known as nodes.
- A variable is assumed to act over each element in a predefined manner, with the number and type of elements chosen so that the variable distribution through the whole body is adequately approximated by the combined elemental representations.
- After the problem has been broken up, the governing equations for each element are calculated and then assembled to give the system equations.
- The equations of a particular type of element for a specific analysis type have a constant format.
- Substituting the nodal co-ordinates, material properties and loading conditions for each of the elements into the general format.

FEA model of a prosthetic knee

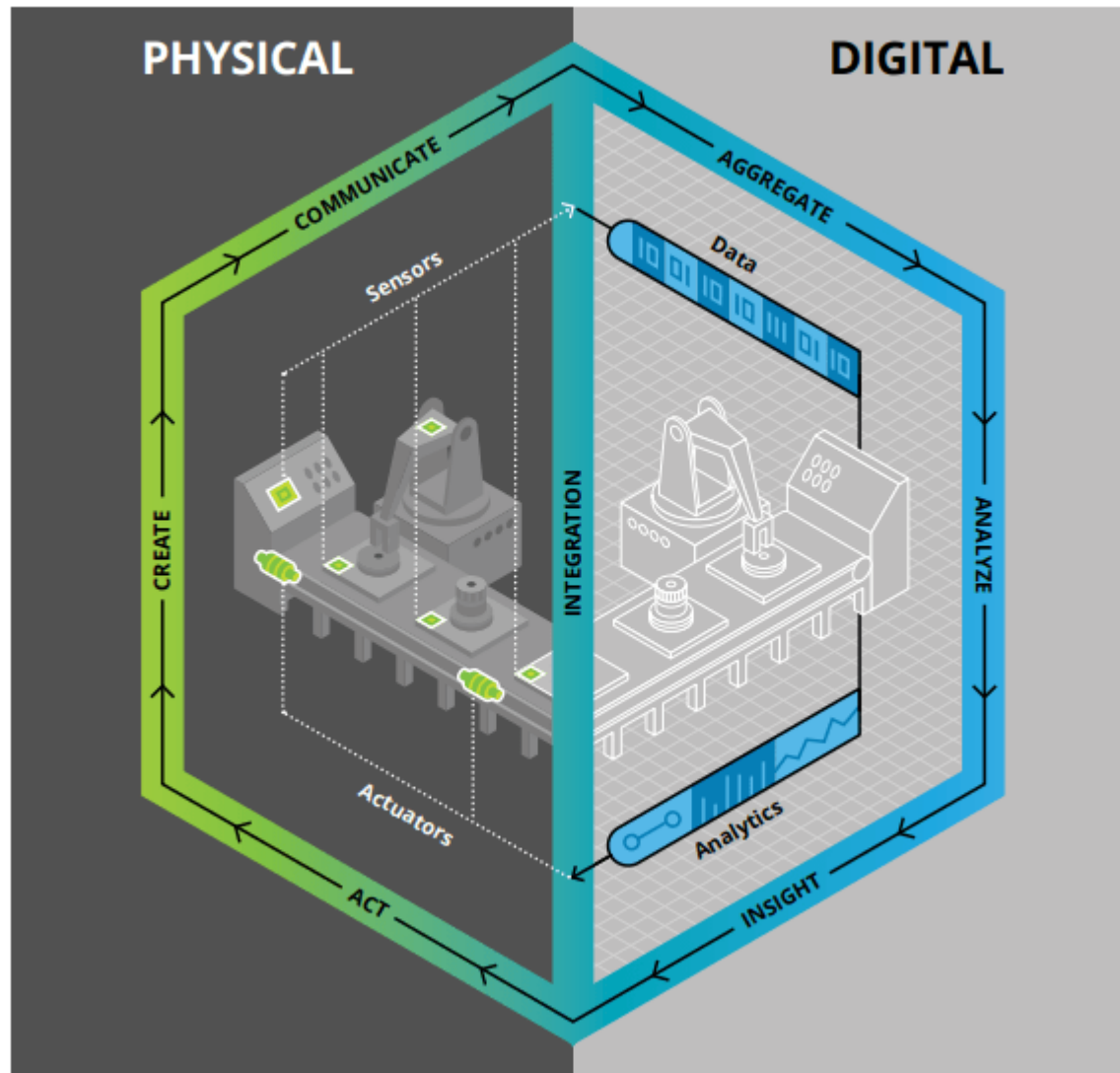


FEA of craniofacial distraction procedures.



Performed by Dr. Daniel Remmeler, Medical School of Wisconsin.

What is a Digital Twin



- Digital twins offer accurate digital representations of physical assets, processes, and the control and automation systems that transform data into actionable insights.
- They provide analysis and decision-making capabilities throughout the design cycle or product lifecycle.
- Digital twins encompass process simulation, data-driven ML models, asset models and connections to real-world data synchronised from multiple sources including operational data, engineering data, maintenance data and financial information.

Using Digital Twins

